Optimal Control of a CSTR with PI Controller: Problem Formulation and Implementation

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Contents

1 Overview

This document describes an **optimal control problem** for a **Continuous Stirred Tank Reactor (CSTR)** with integrated **PI (Proportional-Integral) controller** for temperature regulation. The system simultaneously optimizes:

- **Reactor design parameters**: Diameter (D_R) and height (H_R)
- **PI controller tuning**: Gain (K_c) and integral time (τ_I)
- **Dynamic control trajectory**: Jacket flow rate (F_J) to regulate reactor temperature

1.1 System description

The CSTR consists of:

- A **reactor vessel** with variable diameter and height, containing an exothermic chemical reaction
- A **cooling jacket** surrounding the reactor to remove heat via heat transfer
- A **PI controller** that manipulates jacket flow rate to maintain reactor temperature at setpoint

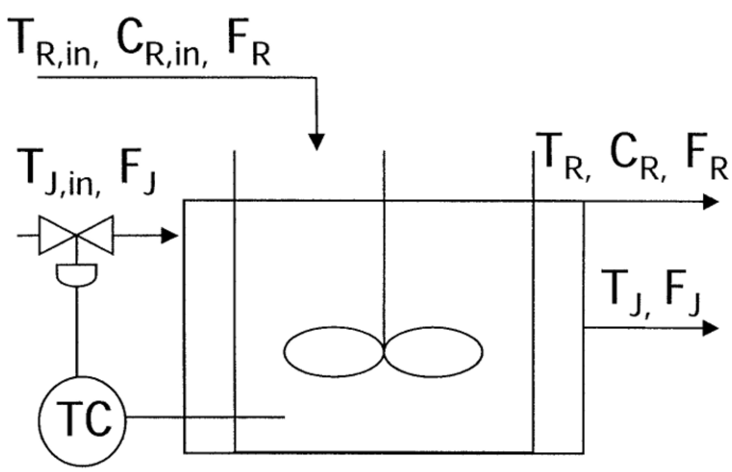

Figure 1: Continuous Stirred Tank Reactor (CSTR) schematic showing reactor vessel with cooling jacket

1.2 State variables

- $C_R(t)$: Reactor concentration [mol/m^3]
- $T_R(t)$: Reactor temperature [K] (controlled variable)
- $T_J(t)$: Jacket temperature [K]

1.3 Control and design variables

- $F_{J,c}(k)$: Jacket flow rate at control step k [m³/hr] (manipulated variable)
- D_R: Reactor diameter [m] (design parameter)
- H_R: Reactor height [m] (design parameter)
- K_c: PI controller gain (tuning parameter)
- τ_I: PI controller integral time [hr] (tuning parameter)

1.4 Objective

Minimize the initial reactor temperature $T_R(0)$ subject to:

- Reaction kinetics (Arrhenius equation)
- Energy balances (reactor and jacket)
- Mass balance (concentration dynamics)
- PI controller dynamics
- Endpoint stabilization constraint (return to setpoint within tolerance)

The problem features a temperature disturbance at $t = 0.1$ hr, and the controller must reject this disturbance while maintaining stability.

2 Notation and Parameters

2.1 Time discretization

Two time grids are used:

- **ODE integration grid**: $t_n = n \cdot \delta_n$, $n = 0, 1, \ldots, N-1$ with $\delta_n = 0.05$ hr
- **Control update grid**: $t_k = k \cdot \Delta t$, $k = 0, 1, \ldots, K-1$ with $\Delta t = 0.1$ hr

Time horizon: $t_f = 1.0$ hr, resulting in:

$$N = 21 \quad \text{(ODE steps)} \tag{1}$$

$$K = 11 \quad \text{(control steps)} \tag{2}$$

$$K_{\text{inc}} = 2 \quad \text{(ODE steps per control update)} \tag{3}$$

2.2 Physical parameters

Table 1: Reaction and heat transfer parameters

Symbol	Description	Value / Units
ΔH	Heat of reaction	-31652 kJ/mol
U	Overall heat transfer coefficient	1893 kJ/(hr m^2 K)
E_a/R	Activation energy / gas constant	8375 K
k_0	Pre-exponential factor	4.08×10^{10} hr^{-1}

Table 2: Fluid properties

Symbol	**Description**	**Value / Units**
ρ	Reactor fluid density	800.9 kg/m^3
C_p	Reactor fluid heat capacity	0.9691 kJ/(kg K)
ρ_j	Jacket fluid density	997.95 kg/m^3
C_j	Jacket fluid heat capacity	1.292 kJ/(kg K)

Table 3: Operating conditions and disturbance

Symbol	**Description**	**Value / Units**
F_R	Reactor flow rate	2.832 m^3/hr
$T_{R,\text{in},0}$	Initial inlet temperature	333.0 K
$\Delta T_{R,\text{in}}$	Temperature disturbance step	+0.5 K
$T_{J,\text{in}}$	Jacket inlet temperature	294.1 K
$C_{R,\text{in}}$	Inlet concentration	10.89 mol/m^3

The inlet temperature undergoes a step change:

$$T_{R,\text{in}}(t) = \begin{cases} 333.0 \text{ K} & \text{if } t < 0.1 \text{ hr} \\ 333.5 \text{ K} & \text{if } t \geq 0.1 \text{ hr} \end{cases} \tag{4}$$

2.3 Variable bounds

Table 4: Bounds on state, control, and design variables

Variable	**Description**	**Lower**	**Upper**
T_R	Reactor temperature	333.0 K	360.8 K
T_J	Jacket temperature	300.0 K	1110.0 K
$F_{J,c}$	Jacket flow rate (control)	20.553 m^3/hr	20.6 m^3/hr
D_R	Reactor diameter	3.0 m	6.0 m
H_R	Reactor height	4.0 m	10.0 m
K_c	Controller gain	-30.0	10.0
τ_I	Integral time	0.1 hr	4.0 hr

Note: $C_R \geq 0$ (non-negative concentration).

2.4 Endpoint tolerance constraint

To ensure the system returns to setpoint (initial temperature) at the end of the horizon:

$$|T_R(n) - T_R(0)| \leq \epsilon_{\text{tol}}, \quad n = N - n_{\text{tol}}, \ldots, N - 1 \tag{5}$$

where $\epsilon_{\text{tol}} = 5 \times 10^{-4}$ K and $n_{\text{tol}} = 2$ (last 2 time steps).

3 Continuous-Time Model

3.1 Reactor geometry

The reactor is a cylindrical vessel with:

$$V_R = \frac{\pi}{4} D_R^2 H_R \quad \text{(reactor volume)} \tag{6}$$

$$A_R = \pi D_R H_R \quad \text{(reactor surface area)} \tag{7}$$

$$V_J = \frac{\pi}{3} D_R H_R \quad \text{(jacket volume)} \tag{8}$$

3.2 Reaction kinetics

The reaction rate follows the Arrhenius equation:

$$k_1(t) = k_0 \exp\left(-\frac{E_a/R}{T_R(t)}\right) \tag{9}$$

3.3 Concentration dynamics

Mass balance for the reactor:

$$V_R \frac{dC_R}{dt} = F_R(C_{R,\text{in}} - C_R) - k_1 V_R C_R \tag{10}$$

Dividing by V_R:

$$\frac{dC_R}{dt} = \frac{F_R}{V_R}(C_{R,\text{in}} - C_R) - k_1 C_R \tag{11}$$

3.4 Reactor temperature dynamics

Energy balance for the reactor:

$$V_R \rho C_p \frac{dT_R}{dt} = F_R \rho C_p (T_{R,\text{in}} - T_R) - \Delta H \cdot k_1 V_R C_R - U A_R (T_R - T_J) \tag{12}$$

Dividing by $V_R \rho C_p$:

$$\frac{dT_R}{dt} = \frac{F_R}{V_R}(T_{R,\text{in}} - T_R) - \frac{\Delta H}{\rho C_p} k_1 C_R - \frac{U A_R}{V_R \rho C_p}(T_R - T_J) \tag{13}$$

3.5 Jacket temperature dynamics

Energy balance for the jacket:

$$V_J \rho_j C_j \frac{dT_J}{dt} = F_{J,c} \rho_j C_j (T_{J,\text{in}} - T_J) + U A_R (T_R - T_J) \tag{14}$$

Dividing by $V_J \rho_j C_j$:

$$\frac{dT_J}{dt} = \frac{F_{J,c}}{V_J}(T_{J,\text{in}} - T_J) + \frac{U A_R}{V_J \rho_j C_j}(T_R - T_J) \tag{15}$$

4 Discrete-Time Model (Backward Euler)

The continuous-time ODEs are discretized using the **backward Euler** method:

$$\left.\frac{dx}{dt}\right|_{t=t_n} \approx \frac{x_n - x_{n-1}}{\delta_n} \tag{16}$$

4.1 Reaction rate constraint

At each ODE time step n:

$$k_{1,n} = k_0 \exp\left(-\frac{E_a/R}{T_{R,n}}\right) \tag{17}$$

4.2 Initial conditions (algebraic equations at $t = 0$)

At steady state ($t = 0$), the time derivatives are zero, yielding algebraic equations:

Concentration:

$$0 = F_R(C_{R,\text{in}} - C_{R,0}) - k_{1,0} V_R C_{R,0} \tag{18}$$

Reactor temperature:

$$0 = F_R(T_{R,\text{in},0} - T_{R,0}) - \frac{\Delta H}{\rho C_p} k_{1,0} V_R C_{R,0} - \frac{UA_R}{\rho C_p}(T_{R,0} - T_{J,0}) \tag{19}$$

Jacket temperature:

$$0 = F_{J,c,0}(T_{J,\text{in}} - T_{J,0}) + \frac{UA_R}{\rho_j C_j}(T_{R,0} - T_{J,0}) \tag{20}$$

4.3 Dynamic constraints (backward Euler, $n \geq 1$)

Concentration dynamics:

$$V_R C_{R,n} = V_R C_{R,n-1} + \delta_n \left[F_R(C_{R,\text{in}} - C_{R,n}) - k_{1,n} V_R C_{R,n}\right] \tag{21}$$

Reactor temperature dynamics:

$$V_R T_{R,n} = V_R T_{R,n-1} + \delta_n \Bigg[F_R(T_{R,\text{in}}(t_{n-1}) - T_{R,n}) - \frac{\Delta H}{\rho C_p} k_{1,n} V_R C_{R,n} - \frac{UA_R}{\rho C_p}(T_{R,n} - T_{J,n})\Bigg] \tag{22}$$

where $T_{R,\text{in}}(t_{n-1})$ accounts for the step disturbance at $t = 0.1$ hr.

Jacket temperature dynamics: The control input $F_{J,c}$ is updated every $K_{\text{inc}} = 2$ ODE steps. Map ODE index n to control index k:

$$k = \left\lfloor \frac{n-1}{K_{\text{inc}}} \right\rfloor \tag{23}$$

$$V_J T_{J,n} = V_J T_{J,n-1} + \delta_n \left[F_{J,c,k}(T_{J,\text{in}} - T_{J,n}) + \frac{UA_R}{\rho_j C_j}(T_{R,n} - T_{J,n}) \right] \tag{24}$$

5 PI Controller Formulation

The PI controller manipulates the jacket flow rate $F_{J,c}$ to maintain the reactor temperature at the setpoint $T_{R,0}$ (initial temperature).

5.1 Control error

At control step k, the tracking error is:

$$e_k = T_{R,0} - T_{R,n(k)} \tag{25}$$

where $n(k) = \min(k \cdot K_{\text{inc}}, N-1)$ maps control index to ODE index.

Initial condition: $e_0 = 0$.

5.2 Incremental PI control law

The control increment is computed using the velocity (incremental) form of the PI algorithm:

$$\Delta u_k = K_c \left[(e_k - e_{k-1}) + \frac{\Delta t}{\tau_I} e_k \right], \quad k \geq 1 \tag{26}$$

Initial condition: $\Delta u_0 = 0$.

5.3 Control update

The manipulated variable is updated by adding the increment:

$$F_{J,c,k} = F_{J,c,k-1} + \Delta u_k, \quad k \geq 1 \tag{27}$$

The control action is saturated by the bounds: $F_{J,c,\min} \leq F_{J,c,k} \leq F_{J,c,\max}$.

6 Endpoint Stabilization Constraint

To ensure the system returns to the setpoint (initial temperature) by the end of the time horizon, we enforce:

$$-\epsilon_{\text{tol}} \leq T_{R,n} - T_{R,0} \leq \epsilon_{\text{tol}}, \quad n = N - n_{\text{tol}}, \ldots, N-1 \tag{28}$$

This constraint applies to the last $n_{\text{tol}} = 2$ time steps.

7 Complete Optimization Problem

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\boxed{
\begin{aligned}
\min_{C_R, T_R, T_J, F_{J,c}, k_1, e, \Delta u, D_R, H_R, K_c, \tau_I} \quad & T_{R,0} \\
\text{subject to:} \quad & \\
& \text{Reaction rate:} \quad k_{1,n} = k_0 \exp\left(-\frac{E_a/R}{T_{R,n}}\right), \quad n = 0, \ldots, N-1 \\
& \text{Initial conditions:} \quad \text{Eqs. (18) to (20)} \\
& \text{Dynamics:} \quad \text{Eqs. (21), (22) and (24)}, \quad n = 1, \ldots, N-1 \\
& \text{PI controller:} \quad \text{Eqs. (25) to (27)}, \quad k = 0, \ldots, K-1 \\
& \text{Endpoint tolerance:} \quad \text{Eq. (28)} \\
& \text{Bounds:} \quad \text{(see Table 4)}
\end{aligned}
}
\tag{29}
$$

8 Problem Characteristics

8.1 Nonlinearity

The problem is **highly nonlinear** due to:

- Arrhenius reaction rate: $k_1 = k_0 \exp(-E_a/(RT_R))$
- Nonlinear coupling between temperature and concentration in energy balance
- Geometric dependencies: V_R, A_R, V_J are nonlinear functions of D_R, H_R

8.2 Multi-timescale structure

Two time grids are used:

- **Fast dynamics**: ODE integration at $\delta_n = 0.05$ hr (21 steps)
- **Slow control updates**: PI controller updates at $\Delta t = 0.1$ hr (11 steps)

This reflects practical implementation where control actions are updated less frequently than state measurements.

8.3 Disturbance rejection

The system is subjected to a step disturbance in inlet temperature at $t = 0.1$ hr. The optimal controller must:

- Maintain temperature regulation despite the disturbance
- Return to setpoint within tolerance by end of horizon
- Respect all state and control bounds throughout the transient

8.4 Simultaneous design and control optimization

The problem simultaneously optimizes:

(1) **Design parameters**: Reactor geometry (D_R, H_R)

(2) **Controller tuning**: PI gains (K_c, τ_I)

(3) **Control trajectory**: Dynamic jacket flow rate ($F_{J,c,k}$)

This is a challenging **integrated design and control** problem, where physical design and control performance are optimized together.

9 Solution Approach

The problem is formulated as a **nonlinear program (NLP)** and solved using the **IPOPT** solver via the **Pyomo** optimization framework.

9.1 Implementation details

- **Modeling framework**: Pyomo (Python-based algebraic modeling)
- **Solver**: IPOPT (Interior Point Optimizer)
- **Discretization**: Backward Euler (implicit time integration)
- **Variables**: N state variables, K control variables, 4 design/tuning parameters
- **Constraints**: Nonlinear algebraic and differential equations

9.2 Solver settings

- Maximum iterations: 3000
- Convergence tolerance: 10^{-6}
- Derivative computation: Automatic differentiation (Pyomo)

10 Results Interpretation

The optimal solution provides:

(1) **Optimal reactor design**:
    - Diameter D_R^* and height H_R^*
    - These determine reactor volume, surface area, and jacket volume

(2) **Optimal PI controller tuning**:
    - Controller gain K_c^* (typically negative for cooling control)
    - Integral time τ_I^* (determines aggressiveness of integral action)

(3) **Optimal control trajectory**:

- Time-varying jacket flow rate $F^*_{J,c,k}$
- Demonstrates how the controller responds to the inlet temperature disturbance

(4) **Optimal state trajectories**:

- Reactor concentration $C^*_R(t)$
- Reactor temperature $T^*_R(t)$ (should track setpoint)
- Jacket temperature $T^*_J(t)$ (adjusts to provide cooling)

10.1 Performance metrics

- **Objective value**: Minimum achievable initial temperature $T^*_{R,0}$
- **Disturbance rejection**: Peak temperature overshoot after disturbance
- **Settling time**: Time to return within tolerance of setpoint
- **Control effort**: Range of jacket flow rate variation

11 Extensions and Future Work

11.1 Economic optimization

The current objective (minimize initial temperature) can be replaced with an economic objective:

$$\min \quad c_{\text{cool}} \sum_k F_{J,c,k} + c_{\text{dev}} \sum_n (T_{R,n} - T_{\text{sp}})^2 + c_{\text{size}}(D_R^2 H_R) \tag{30}$$

where:

- c_{cool}: cost of cooling fluid
- c_{dev}: penalty for temperature deviation
- c_{size}: capital cost of reactor size

11.2 Model Predictive Control (MPC)

The PI controller can be replaced with MPC for improved performance:

- Receding horizon optimization
- Explicit constraint handling
- Predictive disturbance rejection

11.3 Reinforcement Learning

The optimal control problem can serve as a benchmark for RL-based approaches:

- State: $s_k = [C_{R,k}, T_{R,k}, T_{J,k}, e_k]$
- Action: $a_k = F_{J,c,k}$ or Δu_k
- Reward: $r_k = -(T_{R,k} - T_{\text{sp}})^2 - \lambda |F_{J,c,k} - F_{J,c,k-1}|$

12 Summary

This document presents a comprehensive formulation of an optimal control problem for a CSTR with integrated PI controller. Key features include:

- **Nonlinear dynamics**: Arrhenius kinetics, energy balances, mass balance
- **Multi-timescale discretization**: Fast ODE integration, slow control updates
- **Integrated design and control**: Simultaneous optimization of reactor geometry and controller tuning
- **Disturbance rejection**: Step change in inlet temperature
- **Endpoint constraints**: Return to setpoint within tolerance
- **Implicit time integration**: Backward Euler discretization
- **Nonlinear programming**: IPOPT solver via Pyomo framework

The problem serves as a realistic benchmark for advanced control strategies including MPC and reinforcement learning, while demonstrating the integration of physical design optimization with dynamic control performance.